Test What the Input Specified: Input-Conditioned Requirements and a Tested Oracle for AAC Sentence Expansion

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***Abstract*—A person using an augmentative and alternative communication (AAC) app typed "Open this pickle jar for me."; the app's LLM sentence expander returned "Please open this pickle jar for me." Whether that edit is a defect depends on what the input specified: here the person supplied the whole sentence, so the expander changed a choice it was given. When the person selects only keywords, by contrast, words such as "not" or "please" may never reach the model, and its guess is a default, not a violation. We turn this distinction into executable requirements that score a commitment only when the input carried it, and into a partial test oracle whose own failure modes we test. On 238 fresh authored families and four models, all three pre-registered contrasts held after Holm correction; for instance, fully typed plain requests gained a politeness marker in 26/320 outputs, polite ones lost it in 6/320, with most additions from one model. The extractor behind these scores agreed with two human annotators on held-out items at $\kappa = 0.91$–0.97 across all seven label fields. Testing the oracle mattered: relaxing a rule to remove false alarms also hid real substitutions, so the affected families are quarantined.**

***Index Terms*—accessibility, AAC, LLM testing, requirements, test oracle, regression testing**

I. Introduction

People who use AAC often compose a message by selecting a few words or symbols, because every selection costs time and effort [1], [2].

LLM keyword expansion turns such a selection into a sentence [3], [2]; some systems also let the person send a sentence directly [4]. Fig. 1A shows what can happen when the feature is given a sentence the person already typed in full: Gemma-3-4B added "Please". Across four models and 238 fresh directive families, plain typed sentences gained a politeness marker in 26/320 outputs (8.12%), and polite ones lost theirs in 6/320 (Fig. 2). The pattern is concentrated, not universal: Gemma added a marker in 20/80 and gpt-6.1-terra in 5/80, whereas Llama and Qwen removed markers more often than they added them.

Our preservation policy is a software requirement, not a claim about anyone's private intent. A sentence without "please" may or may not reflect a deliberate choice; the requirement is only that the system not change typed wording unless the person asks for rewriting. Adding "please" also changes how a message reads: politeness suggestions shift how writers phrase messages and how they perceive their social relationship [5].

The opposite case looks similar but is not a defect. The keywords `feel well` do not say whether the person feels well; Qwen wrote "I feel well." (Fig. 1B). No model can be blamed for that guess, but the interface decides who must pay an extra selection to say "not". Given `[NOT] feel well`, the same model wrote "I don't feel well." A developer needs to know which case a failed reference match is: a choice the input never carried calls for a better input channel; a choice the model received and changed calls for a preservation check; a fully typed sentence may need no generation at all.

We treat this distinction as an accessibility requirement that can be specified, tested and regression-checked. We contribute (1) *input-conditioned requirements* (§III) that score a commitment only when the visible input supplied it, run over controlled input channels of minimal-pair families; (2) *a partial oracle that is itself tested* (§IV), including a defect that our own repair introduced, with a passed human-agreement gate; (3) *evidence and developer decisions* (§VI–VII): pre-registered contrasts on fresh items, a model- and encoding-specific regression, a corpus transfer that exposed a defect in our own input builder, and a copy baseline for typed sentences. We ran no study with AAC users and make no claim about benefit, effort or preference.

II. Related Work

Testing LLM systems: CheckList organizes behavioral tests into minimum-functionality, invariance and directional tests [6]; Cho et al. catalogue metamorphic relations for LLM-based NLP [7]; COLLIE tests compositional generation constraints [8]. Sensei combines a user simulator with a DSL for assertions and metamorphic relations [9]. Extracting fields, asserting properties and relating test cases are therefore established; we claim no new testing technique. LLM judges show position, verbosity and self-preference biases [10], [11], similarity metrics can miss negation [12], and LLM politeness judgments agree with each other more than with humans [13]. We therefore restrict the judge to extraction of explicit markers and test it. Constraint order affects multi-constraint instruction following [14], which bears on our composition result.

Preservation during rewriting: Qu et al. condition style transfer on the source dialog act and evaluate act retention [15]; preserving a speech act while rewriting is thus not new. LLMs favor particular politeness strategies in free production [16], and rewriting can keep semantic similarity

	Actual visible input ⇒ stored output (`typed:` the person's full sentence; otherwise keywords)	Model, family, output ID	Rate in the example's stratum
A full input	*typed:* `Open this pickle jar for me.` ⇒ "Please open this pickle jar for me."	Gemma-3-4B, W-C r2plt-015/B, `968b8cae`	plain sources gained a marker in 26/320 (this model 20/80); polite sources lost it in 6/320
B channels	bare `feel well` ⇒ "I feel well." +word `feel well today` ⇒ "I feel well today." +chip `[NOT] feel well` ⇒ "I don't feel well." full *typed:* `I'm not feeling well today.` ⇒ "I'm not feeling well today." hidden source: "I'm not feeling well today."	Qwen2.5-7B, W-C r2pol-001/A, `b8710d19`, `c2892e27`, `3e068649`, `f80a4997`	bare outputs affirmative 317/319 (directional; an *Assumption*, not a failure); `[NOT]` targets negated 315/320
C question cue	`[?] parcel signature` ⇒ "Can you sign for this parcel?" hidden source: "Does the parcel from the bank need a signature?"	Llama-3.1-8B, W-C r2act-049/B, `7d7ecf9f`	`[?]` targets not kept as questions 58/208; of these, became requests 42/208

D guard repair	`[PLAIN] let finish sentence`
	"Please let me finish my sentence." → "Let me finish the sentence."
	`[ASK] [YOU] [PLAIN] wear headphones`
	"You wear headphones." → "Ask you to wear headphones."
		Gemma-3-4B pol2-14/A; Llama-3.1-8B pol2-07/B; discovery set, P0→G1	scored repairs 63/80, 17 remain; the second passes the scorer but is a fragment (prefix "Ask you" in 4/63 repairs, post hoc lexical count)

E checker	*typed:* `Yes, you can borrow my car this weekend.` ⇒ "Yes, you can borrow my car this weekend."	Qwen2.5-7B, discovery act-16/B, `b6b079ac`	the model copied the sentence; the original checker failed 14/2,292 such identical outputs, all this sentence (family now quarantined)

Fig. 1. Selected stored outputs, verbatim (11 strings re-read from the run files). Each example is the first event in a declared order (A, C: minimum SHA-256 over the stored event key; B: first negated polarity target by family, variant and model order; D: first and second failure→pass in the guard's hash order; E: first raw-oracle failure on an unchanged sentence); quarantined families are excluded except in E. Selection is conditioned on the displayed event, so the examples are not typical; the right column gives the rate in each stratum. A–C are W-C (fresh items, prompt P0), D–E the discovery set. A *hidden source* was not visible to the model. Full rules and IDs: `ideas/cycles/c2/story_evidence.json`.

while erasing pragmatic markers, which preservation prompts reduce but do not remove [17]. Our focus is narrower: whether a marker the person supplied, or explicitly did not supply, survives.

LLMs in AAC: Systems studied with AAC users [1] expose intent controls such as intent buttons [18], intention choices for people with aphasia [19] and stance tags [20]. Lacosse et al. keep bracketed function words such as "not" [21]; Choudhury et al. test whether affect cues reach the output [22]; Gorenshtein et al. benchmark intent drift under decoder errors [23]; R-SPEAK was co-designed with people with aphasia and scores faithfulness [24]; a wearable system lets users choose direct output or expansion [4]. AAC generation therefore already has controls, optional expansion and fidelity evaluation. We add a conditional specification of *which* commitments are testable under each input, and a test of the oracle. Accessibility engineering spans the life cycle [25]; conformance checks cannot see whether generated text keeps the user's meaning.

III. Input-Conditioned Requirements

Families and channels: A *family* is a minimal pair of authored source messages that share content words and differ on one *critical* dimension: polarity, agent (who acts), certainty, speech act (question, request or statement), politeness, or time/quantity. Each variant is rendered through five channels, shown for one family in Fig. 1B: *bare* keywords; +*word*, which adds one noncritical content word; +*chip*, which adds the critical cue as a bracketed tag (`[NOT]`); +*all*, with every tag the variant needs; and *full*, the source sentence itself. Every variant needs at least one tag, so the unmarked reading gets a counter-tag (e.g., `[PLAIN]`) and neither variant passes for free. Tags are a test device for supplying a cue, not a proposed interface.

Requirements: For output y from input x, let $L_d(y)$ be the extracted value on dimension d and $C(x)$ the commitments that x carries. **R1** (supplied cue): if $C(x)$ fixes d to c, $L_d(y)$ must be admissible for c. **R2** (full input): if x is the full source, each scored explicit value is kept, including the *absence* of a politeness marker. **R3** (ambiguity): if $C(x)$ does not fix d, $L_d(y)$ is not pass/fail; a committed value is logged as an *Assumption*, and its direction is reported, because a systematic default decides which reading is free. **R4** (composition): adding a compatible cue for e must not break a satisfied value on $d \neq e$. **R5** (input builder): a test input that claims to supply commitment c must actually display a cue for c; otherwise failures cannot be attributed to the model (§VI-E).

Verdicts are separate checks: Loss records that the tested channel does not fix the value; *Violation* that a supplied value was not kept (R1/R2); *Assumption* that the output committed to a value the input left open (R3). They are not exclusive: "I feel well." from `feel well` is both a Loss (the channel never carried the negation) and an Assumption (the output chose affirmative), and no Violation. A reference-fidelity score would instead fail it against "I'm not feeling well today.", charging the model for what the interface omitted; a similarity score would pass Fig. 1A, although the input supplied the plain wording. With balanced pairs, a bare output matches at most one variant, so 50% paired match is a design ceiling under

TABLE I
ORACLE EVIDENCE ON AUTHORED INPUTS (WILSON 95% INTERVALS). THESE ARE CHECKS ON CLEAN SENTENCES, NOT ERROR RATES ON MODEL OUTPUTS.

Check	Result	95% CI
Identity: unchanged source, false alarms	0/120	[0.0, 3.1]%
Scoped single-value mutations detected	120/120	[96.9, 100.0]%
Keyed challenge mutants detected (gate)	22/24	[74.2, 97.7]%
Keyed challenge controls, false alarms	0/24	[0.0, 13.8]%
acceptance group, relaxed rule	4/6	[30.0, 90.3]%
all groups, without the relaxation	24/24	[86.2, 100.0]%

ambiguity, not a target.

Executable checks: MR2 (cue restoration) and MR3 (full preservation) are single-output contract checks of R1/R2 on cue and full channels. MR4 relates a single-cue output to a two-cue output and counts pass→fail transitions of the protected value (R4). MR1 compares bare with +word; both leave d open, so a changed label is a default-sensitivity diagnostic and a length control, not a defect. MR0 tests the oracle: an unchanged source must pass. Verdicts are computed in code from labels, the item specification and the channel, so a rule change re-scores all outputs without model calls.

IV. A PARTIAL ORACLE AND ITS FAILURE ENVELOPE

The oracle decides only whether a fixed set of explicit, text-level values is kept. It does not judge fluency, appropriateness, referents or whether anyone would accept the sentence.

Extraction as run: An extractor (gpt-6.1-sol, a different model from the GPT model under test) answers the family's critical question and five general questions. Critical and general extraction see the output only; added-content extraction also sees the visible input. Two prompt variants (J1, J2) differ in wording and option order. Critical disagreements are resolved per run. On the discovery set, a cross-family arbiter (claude-opus-5-5) decides them, otherwise the label is unknown; the arbiter may see the visible input, which in full mode reveals the source. On W-C, the arbiter resolved 259 rows (177 units) after outcomes were known, from an output-only view; the pre-arbitration analysis, which masks them as unknown, is reported beside it (§VI-A). W-D uses a J1 fallback on 591 of 40,960 scored rows. General-dimension ties are broken by a rule-based extractor that the guard in §VII also uses, which limits their independence.

Testing the oracle: Table I lists the evidence. Each authored test source was injected as its own full-channel output (identity) and with one hand-written single-value mutation, frozen before judging. A second, hand-written challenge set has four groups of six controls and six one-feature mutants, keys hashed before the first judge call; its pre-specified gate ($\geq$22/24 mutants detected, $\leq$1/24 false alarms) passes when pooled, with no slack.

A defect, a repair, and its blind spot: Before these injections, MR0 on stored model outputs found that the original oracle failed 14 of 2,292 reproductions of the source, all one sentence, "Yes, you can borrow my car this weekend.", coded

TABLE II
HUMAN GATE ON THE DISCOVERY EXTRACTION INVENTORY. α: KRIPPENDORFF, TWO ANNOTATORS ON THE CORE. κ: EXTRACTOR VS. HUMAN REFERENCE ON HELD-OUT ITEMS, 95% CI FROM 1,000 ITEM BOOTSTRAPS. WEIGHTED, ALL-SAMPLE AND CHALLENGE VIEWS: ARTIFACT.

Field	α	held-out κ [95% CI]	n
critical value	0.97	0.95 [0.90, 0.98]	119
polarity	0.95	0.96 [0.83, 1.00]	121
speech act	1.00	0.94 [0.88, 0.99]	118
agent	0.97	0.91 [0.84, 0.97]	117
certainty	1.00	0.91 [0.83, 0.97]	119
politeness	1.00	0.97 [0.91, 1.00]	121
added content	0.98	0.91 [0.81, 0.98]	107

statement against the target *acceptance* (Fig. 1E). The model had done its job; the checker demanded a distinction the text need not force. We relaxed the rule so that *statement* satisfies an acceptance target. The keyed challenges then showed the cost: the relaxed oracle detects 4/6 acceptance mutants and misses both statement substitutions ("You borrowed my car this weekend."), whereas without the relaxation it detects 24/24. Removing a false alarm removed fault sensitivity. We therefore quarantine both refusal/acceptance families from every primary result (leaving 58 of 60 discovery families) until a predicate passes both identity and mutation tests.

Human gate: Two annotators coded 187 items with the extractor's codebook and its output-only view: a 100-item core coded by both (76 sampled outputs, 24 challenge sentences) and 87 further outputs coded by one. Labels were converted from another format; the files carry no per-item timing. The reference is the plurality label; core disagreements (at most 3 per field) are excluded, not adjudicated. The pre-specified gate ($\kappa \geq 0.6$ on held-out items) passes on all 7 fields (Table II); the lowest lower bound is 0.81. Recomputing each sampled unit's verdict from human labels, Violation verdicts agree on 96/99 primary cue-channel units ($\kappa = 0.92$): the judge flags 23, the humans 24, with 1 judge-only and 2 human-only positives.

Humans caught a shared machine error: On "I can't wait to go to the party." both judge prompts, the rule-based extractor and two further machine coders answered *negated*; both annotators answered *affirmative*, as keyed. The machines agreed perfectly there and were wrong together, so machine agreement is not validation. Humans did not resolve every departure: both read "a couple of painkillers" as *two*, as the machines did, against our frozen key *unclear*; we keep the key and report the disagreement. On critical labels against the keys, the annotators score 23/24 and 22/24, the judge 22/24. The gate validates extraction of explicit markers on the discovery inventory; it does not validate repaired text, W-B readings or every new W-C/W-D boundary case.

V. EVIDENCE POPULATIONS

All items are authored or crowd-written, not messages from AAC users. *Discovery*: 60 families, 10 per dimension, of which 58 are primary after quarantine; prompts and codebook were developed on separate pilot and development families.

TABLE III
W-C REGISTERED CONTRASTS (PROMPT P0, 238 FAMILIES, FOUR MODELS). PAIRS ARE OBSERVED/PLANNED; CLUSTERS ARE RECORDED SEED/CONSTRUCTION CLUSTERS (EFFECTIVE NUMBER IN PARENTHESES); HOLM OVER THE THREE ONE-SIDED TESTS.

Contrast (all prompt P0)	Estimate [95% CI]	n; clusters	Holm p
P1 speech act kept: tag minus +word	+31.41 pp [25.00, 37.94]	624/624; 55 (33.1)	.0003
P2 full input: politeness added minus removed	+6.25 pp [2.50, 10.00]	320/320; 80	.0009
P3 bare polarity: affirmative share	99.37% [98.42, 100.00]%	319/320; 79 (78.0)	.0003

W-C (confirmatory): 240 new families (80 each for polarity, speech act and politeness), frozen and pre-registered with three contrasts; 238 are analyzed after a pre-generation exclusion of two speech-act families with a codebook conflict. *W-D* (composition): 80 physical scenarios crossing `[YES]`/`[NOT]` with `[I]`/`[YOU]` under two encodings of the constant statement cue, two greedy draws each. *W-B* (transfer): 160 crowd-written fictional AAC messages from AACText [26], 80 random and 80 enriched for rare cues, never pooled; 139 have a restored-cue channel. Three open models run locally with 4-bit quantization (Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, Gemma-3-4B-it) and one API model (gpt-6.1-terra). P0 asks for one sentence (greedy); P2 asks for three suggestions with different meanings; S3 draws three P0 samples at temperature 0.7. Discovery intervals use 2,000 whole-family resamples, W-C 10,000 resamples of recorded seed/construction clusters, W-D 10,000 resamples of physical scenarios, W-B message resamples; all are pointwise 95% percentile intervals over a fixed model roster, and zero observed events do not establish zero risk.

VI. RESULTS

A. Registered contrasts on fresh items (W-C)

Table III gives the three registered contrasts, all on prompt P0 (the names P1–P3 are contrast labels, not prompts). All 3 pass Holm correction. Confirmation is qualified in two ways. First, 48 cached requests, 12 of them in primary contrasts, predate the final item freeze; excluding every affected family leaves P1 at 29.56 pp [23.06, 36.27]. Second, the speech-act contrast relies on post-hoc arbitration of judge disagreements; with those labels masked as unknown, P1 is 31.65 pp [25.22, 38.19] on 613 pairs. P2 and P3 contain no arbitrated rows.

B. Full-input preservation (R2)

In the discovery set, plain directives gained a marker in 9/40 full-input cases; on fresh W-C items the rate is 8.12% [5.31, 11.25]% (26/320), against 1.88% removals (Fig. 2). The direction replicated and the magnitude shrank. The pooled effect is driven by Gemma (20/80 additions vs. 1/80 removals) and gpt-6.1-terra (5/80 vs. 0/80); Llama (1/80 vs. 2/80) and Qwen (0/80 vs. 3/80) go the other way. This is a model-specific

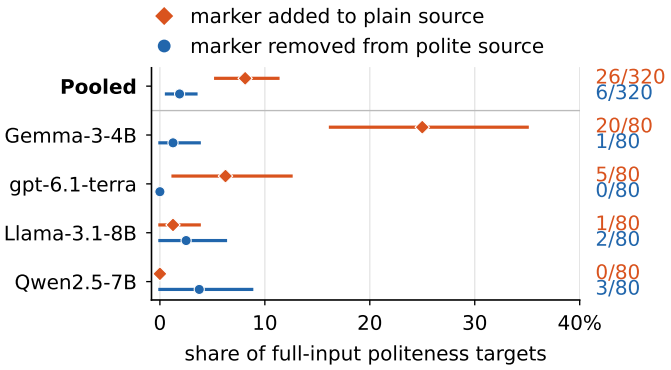


Fig. 2. W-C, full input (the person's sentence is visible to the model), P0: politeness marker added to a plain source vs. removed from a polite one, per model and pooled; 95% cluster-bootstrap intervals; counts at right.

regression to test for, not a property of LLMs. Polarity was kept in 99.84% of full-input targets and speech act in 99.20%.

C. Channel attribution (R1, R3)

Fig. 3 separates the two kinds of evidence. Under bare input, 317/319 directional polarity outputs were affirmative (317/320 counting one unclear output); these are Assumptions, not failures. Once supplied, a negation tag worked: 315/320 negated targets were negated with `[NOT]`. Speech act did not reach full-input fidelity: one tag kept the act in 73.24% of targets against 99.20% with full input, a paired gap of −25.96 pp [−30.36, −21.53]. Question tags failed in 58/208 targets (27.88% [18.86, 37.50]%); 42 of these became requests, as when `[?] parcel signature`, from a question about whether a signature is needed, became "Can you sign for this parcel?" (Fig. 1C). Focal checks overstate preservation when several cues are present: with all tags, joint preservation of every supplied predicate was 66.51% for speech act (focal 74.84%) and 75.74% for politeness (focal 87.19%). Asking for three different meanings (P2) produced both polarity readings in only 25/320 bare sets, against 2/320 for three samples. Diversity can also cut the wrong way: under full input, 21/639 P2 polarity sets contained the *other* source sentence verbatim (2/639 for S3; exploratory, judge-free string match).

D. Composition is a per-model, per-encoding regression (R4)

In W-D (Fig. 4), adding `[YOU]` to `[NOT] [STATEMENT] lock door` turned Llama's "I will not lock the door." into "You lock the door." Across scenarios, Llama lost an already-kept negation in 55/640 transitions, in 30 of 80 scenarios, against 0/160 disagreements when the identical single-cue input was regenerated: 8.59 pp [6.25, 11.09] above that floor, and 7.03 pp [4.38, 9.84] above the floor for two-cue inputs (5/320). All 55 damaged transitions were in the NOT×YOU cell (55/160, against 0/480 in the other three cells; exploratory decomposition), and all lack the checked negation markers. The other three models showed 0 such transitions under this encoding. Under the alternative `[TELL]` encoding Llama kept negation, but Qwen lost the agent in 84/640 transitions (44

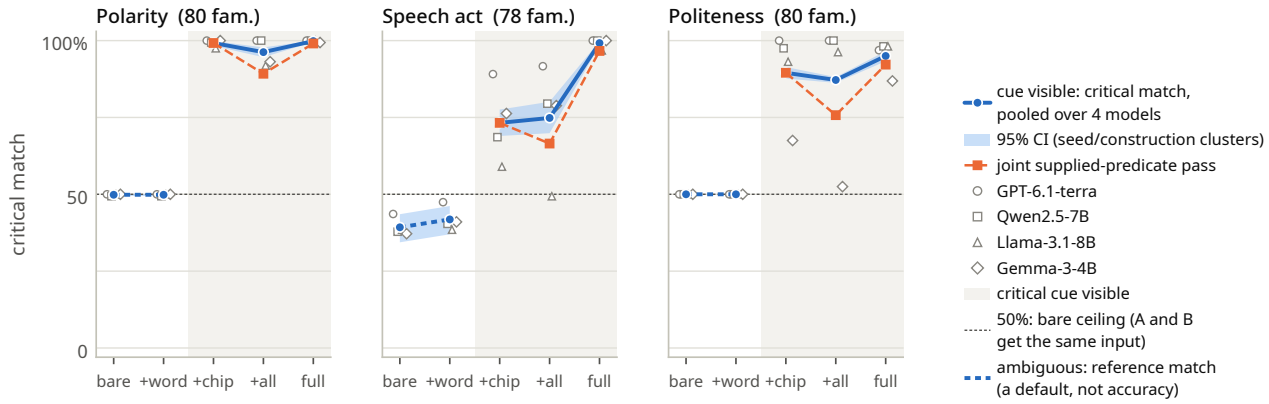


Fig. 3. W-C critical-value match by channel (P0, four models pooled; markers: individual models). Left of the shading, input is ambiguous: match is the chosen default, bounded by a 50% design ceiling, not accuracy. In the shaded region the cue is visible and match measures preservation. Orange squares also require every other supplied predicate to be kept.

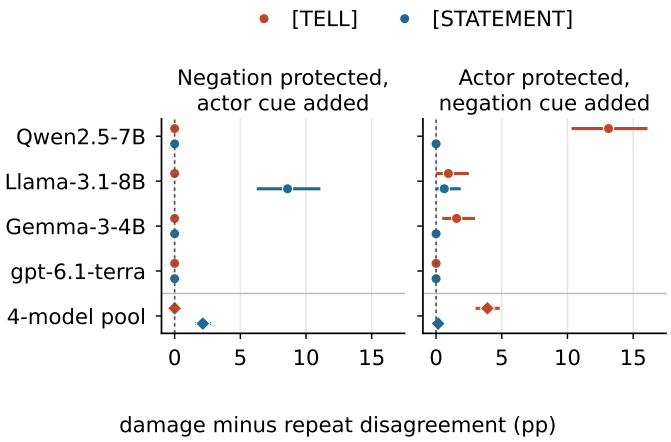


Fig. 4. W-D: protected-value pass→fail transitions when a second cue is added, minus disagreement when the identical single-cue input is regenerated (640 transitions and 160 repeat pairs per point; 95% CIs over 80 scenarios). Two encodings of the same statement cue.

scenarios), and in 0 under `[STATEMENT]`; some of those outputs literalize the tag, and their agent labels depend on a codebook rule that the judges did not apply consistently. Neither encoding is safe for every model, and constraint order is an untested alternative explanation [14]. The engineering consequence is narrow: composition checks must be run per model and per encoding, with a repeat-draw floor (Fig. 4).

E. Corpus transfer and an input-contract defect (W-B)

On AACText messages compressed to four content words, on dimensions the panel judged ambiguous, outputs took the unmarked reading in 84.4% [80.6, 88.2]% of cases (random sample) and 77.4% (enriched). A strictly negated source was kept in 0/32 random and 2/48 enriched outputs. Restoring the focal cue raised source match by +48.9 and +37.1 pp. Auditing all 139 restorations against R5 exposed a defect in our own input builder: 7 displayed a chip that contradicts the stored source category, e.g., `[MAYBE]` for a plain assertion, after a fallback in the builder. In the five `[MAYBE]` cases, all 20 outputs followed the visible chip and failed the source, which a source-fidelity score counts as model failures. Scored against what the input actually displayed, general-predicate fidelity was 96.1% [93.1, 98.3]% (random) and 88.0% [82.6, 92.7]% (enriched); 16 further restorations stay unknown for source attribution (unclear or disputed references, ambiguous words). Full-source category mismatches (4.9% and 5.7%) partly reflect differences between the source panel's and the judge's codebooks. W-B readings are machine-assessed; its 20-message human audit has not been completed.

VII. From Requirement to Developer Decision

Table IV shows how each requirement becomes a check, what the check found, what a developer could decide, and what the evidence does not cover. The full-mode row is the simplest lesson: if the person can say "this is my sentence", the requirement is met by not generating at all; whether full mode is offered and discoverable is a product decision. The guard row shows that the same checks can run at run time; its benefits, residual failures and displaced choices are evaluated separately in companion work and are not claimed here. The oracle and input-builder rows show why test infrastructure needs its own tests: our first oracle repair silently weakened the oracle, and our input builder silently supplied the wrong cue.

Recommendations: Write each requirement conditional on the input and per dimension: "preserve meaning" is not testable, "if the input carries `[NOT]`, the output is negated" is. Test with minimal pairs, keep defaults separate from failures, and report joint as well as focal preservation. Treat added politeness on typed text as a failing test and make style assistance opt-in. Give the oracle identity and keyed-mutation tests and re-run them after every relaxation; validate generated test inputs against their specification. Gate composition per model and encoding, against repeat draws.

TABLE IV
REQUIREMENT → EXECUTABLE CHECK → DEVELOPER DECISION → FAILURE ENVELOPE. MACHINE-SCORED; GUARD RATES ARE DISCOVERY CUE-CHANNEL TARGETS (58 FAMILIES).

Requirement	Check and finding	Developer decision (evaluated)	Failure envelope
R2: a typed sentence keeps its scored values	MR3: W-C plain sources gained politeness in 26/320, mostly one model	Route an explicitly selected full mode to exact copy: 116/116 source identities pass, zero model calls (a baseline, not a finding)	Applies only when the person declares full mode; does not detect whether keywords already form a sentence.
R1: a supplied cue is honored	MR2: speech-act tag 73.24% vs. full 99.20%; discovery strict pass 94.3%	Runtime guard: rule-based check against supplied cues; on a flag, regenerate with feedback up to two times. Feasibility on discovery items: repairs 63/80 failures, strict pass 94.3%→98.8% on 1,392 targets, 0.104 extra calls/input; stripping markers alone repairs 34/80	17 failures remain; repairs are scored predicates, not fluent speech (Fig. 1D); monitor and evaluator share a rule-based component; not run on W-C.
R4: added cues keep protected values	MR4 with repeat floor: Llama with `[STATEMENT]`: 55/640	Gate each model×encoding pair separately; repeat draws before release	Encoding-specific; no universally safe encoding found.
R5: test inputs display the claimed cue	Audit: 7/139 restorations contradict their source	Validate every built input against its specification before generation	Historical requests not rebuilt; affected rows withheld from attribution.
MR0: an unchanged source passes	Identity on stored outputs: 14 false failures, one sentence	Relax acceptance; re-test with keyed mutants; on failure, quarantine	Refusal/acceptance predicate unresolved.

VIII. THREATS TO VALIDITY

Construct. Labels are coarse, explicit categories; "plain" means no explicit marker, and content, referents, negation scope and certainty degree are only partly covered. A passing check is not a correct sentence (Fig. 1D). *Oracle.* All rates depend on an LLM extractor; the human gate covers explicit markers on the discovery inventory with two annotators, plurality references and excluded ties, not subtler readings, W-B readings, repaired text, or every W-C/W-D boundary. The planned 20-item W-C human source check was not completed (0 returns). *Chronology.* W-C has early cached requests and post-hoc arbitration; W-D's item and codebook freeze is not independently established and uses a J1 fallback. *Selection and scale.* Four fixed models; examples were selected after outcomes by declared rules; discovery has ten families per dimension, W-C covers three dimensions. *Heterogeneity.* The politeness and composition effects are model- and encoding-specific. *Cost.* Selection counts omit reading, navigation and fatigue [27]. *No users.* We cannot speak to benefit, agency or whether anyone wants a given output, and we assume no particular language style of any group of AAC users [28], [29].

IX. CONCLUSION

Even a sentence the person typed in full can come back changed. Conditioning requirements on what the input carried separates that violation from the guesses an interface forces when it collects less, and turns both into executable checks with a developer decision attached. On fresh items, the registered contrasts held, with smaller and model-specific effects than in discovery. Testing the test infrastructure mattered as much as testing the models. Families, requirements, harness, oracle tests, stored outputs and analysis code are available at https://github.com/1unarzDev/aac_voice_preservation.

ETHICS

No data were collected from AAC users; AACText is public under CC BY 4.0. Two human annotators labeled model outputs under a fixed codebook.[PENDING: institutional determination; annotator recruitment and compensation]

ACKNOWLEDGMENT

[PENDING: IEEE AI disclosure: name the AI system(s) that drafted text, figures and code, the sections affected and the level of use (VENUE.md). The models under test, the extractor and the arbiter are research instruments described in Sections IV–V.]

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